

Diachronic Generative Syntax: from 2 Factors to 3

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The plan

Session 1 (Wednesday 24 June)

Diachronic generative syntax in the 2 and pre-3 Factors era (1979 - ±2010)

- an **acquisition**-centred approach to variation and change
- a **parameter**-centred approach: 'classic' and post-1980s parameters
- problems with UG-given (Factor 1-based) parameters
- **pathways of change**: grammaticalisation and cycles

throughout: "lurking" precursors to Factor 3

Session 2 (Thursday 25 June)

Diachronic generative syntax in the 3 Factors era (since ca. 2010)

- an **emergentist** approach to acquisition, variation and change: the Maximise Minimal Means (MMM) model
 - a role for children and for adults
- emergent parameters and parametric change
- grammaticalisation, pragmaticalisation, cycles
- exaptation
- I-contact: simplification and complexification (the relevance of sociosyntax)

Reminder: The Factors

- Recall the factors

(1) The classic generative “algorithm”

Universal Grammar (UG) + Input (PLD) → Adult Grammar (I-language)

rich UG

2 factors

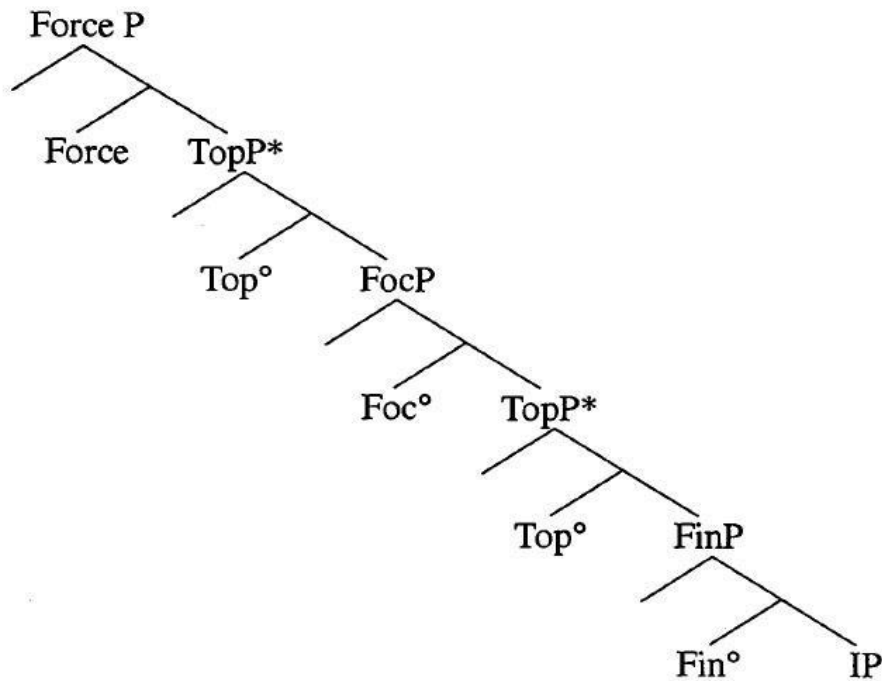
(2) The minimalist generative “algorithm”

UG + PLD + general cognitive factors → Adult Grammar (I-language)

impoverished UG > language structure = largely **emergent**

3 factors

3 Factors and 2 new ideas



AdvP_{speech act} (frankly,...)
 AdvP_{evaluative} (oddly,...)
 AdvP_{evidential} (allegedly,...)
 AdvP_{epistemic} (probably,...)
 AdvP_{past/future} (then,...)
 AdvP_{necessity} (necessarily,...)
 AdvP_{possibility} (possibly,...)
 AdvP_{habitual} (usually,...)
 AdvP_{delayed} (finally,...)
 AspectP_{predispositional} (tendentially,...)
 AdvP_{repetitive} (again,...)
 AdvP_{frequentative} (frequently,...)
 AdvP_{volition} (willingly,...)
 AdvP_{celerative} (quickly,...)
 AdvP_{anterior} (already)
 AdvP_{terminative} (no longer,...)
 AdvP_{continuative} (still,...)
 AdvP_{continuous} (always,...)
 AdvP_{retrospective} (just,...)
 AspectP_{proximative} (soon,...)
 AdvP_{durative/ progressive} (briefly,...)
 AdvP_{prospective} (imminently,...)
 AdvP_{obligation} (obligatorily,...)
 AdvP_{frustrative} (in vain,...)
 AdvP_{completive} (partially,...)
 AdvP_{manner} (well,...)
 Verb

Mood_{speech act}
 Mood_{evaluative}
 Mood_{evidential}
 Mod_{epistemic}
 Tense_{past/future}
 Mod_{necessity}
 Mod_{possibility}
 Aspect_{habitual}
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 Aspect_{durative/progressive}
 Aspect_{prospective}
 Mod_{obligation}
 Aspect_{frustrative}
 Aspect_{completive}
 Voice_{passive}
 Verb

3 Factors and 2 new ideas

- Chomsky (2005): *Three Factors in Language Design*

(2) The minimalist generative “algorithm”

UG + PLD + general cognitive factors → Adult Grammar (I-language)

impoverished UG > language structure = largely **emergent**

- **“poor” UG:**

- parameters are not hard-wired (UG-given)
- What is in UG?
 - Merge (External and Internal)
 - ? Agree
 - ?? a universally given inventory of formal features ([F]s) from which specific grammars make a one-time selection (Chomsky 2001: 10)
 - a minimal inventory of innate formal features, which serve as the basis for emergent features

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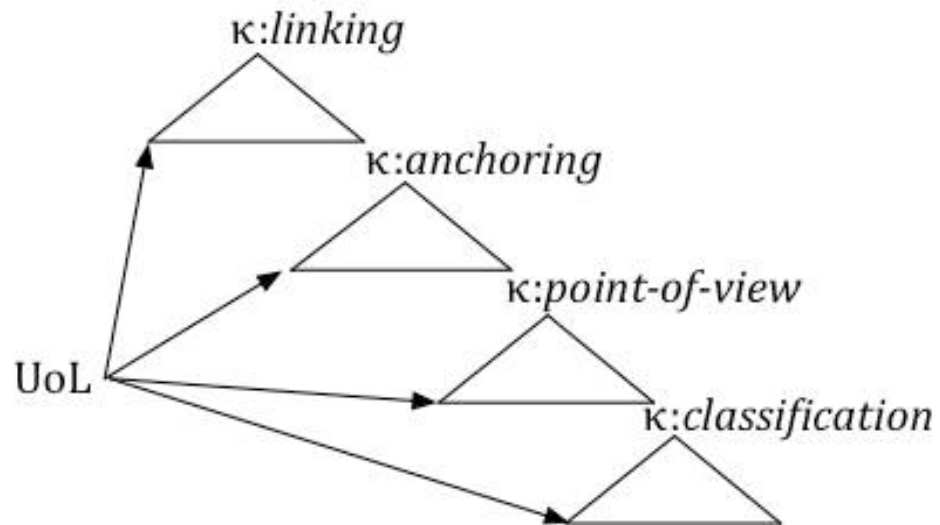
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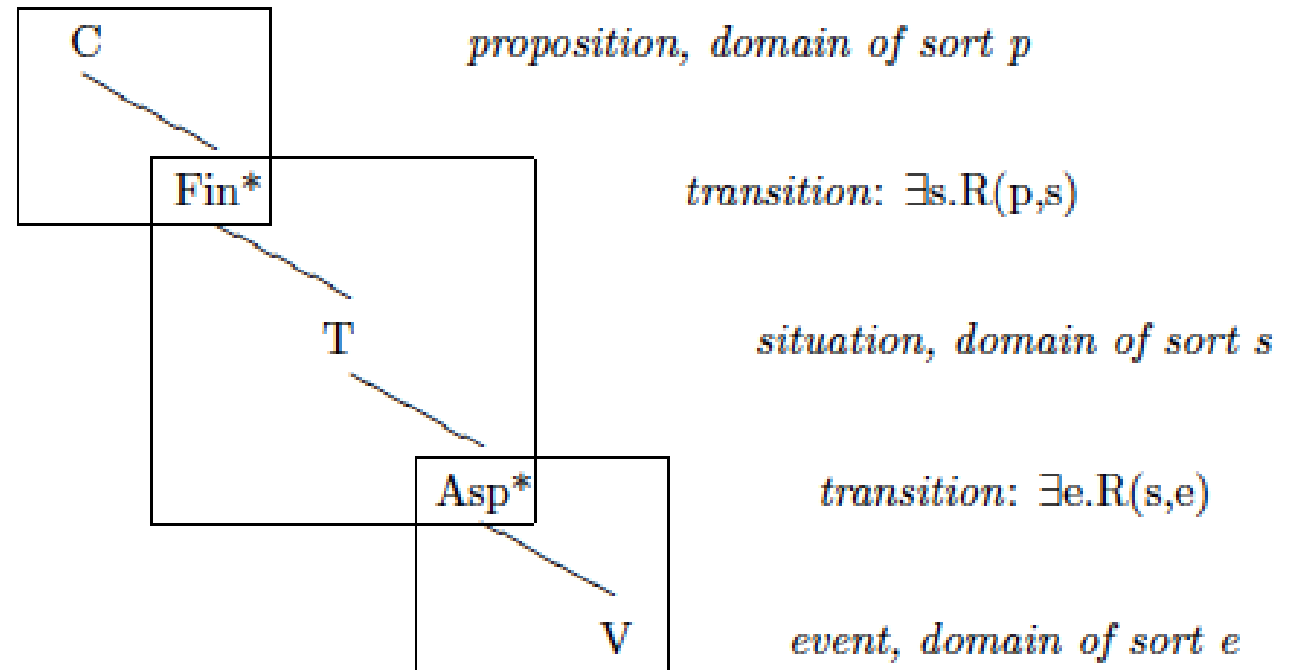
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3 Factors and 2 new ideas

Wiltschko (2014)s Universal Spine



Ramchand & Svenonius (2014)



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 - a formal feature [F] template

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3 Factors and 2 new ideas

- **Emergentism** (philosophy, biology, systems theory) is typically ***contrasted*** with Generativism in Linguistics:

Nativism vs Emergentism



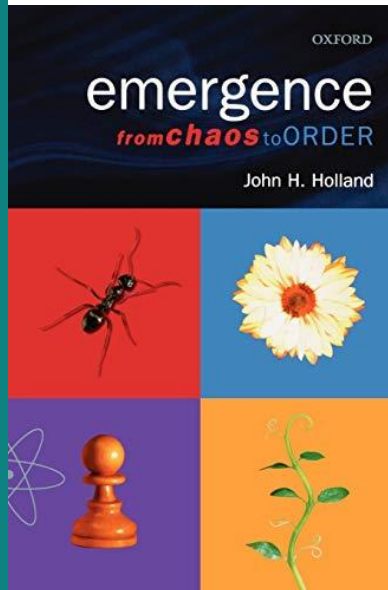
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Nativism vs Emergentism

Language Is a Complex Adaptive System: Position Paper

The “Five Graces Group”, Clay Beckner, Richard Blythe, Joan Bybee,
Morten H. Christiansen, William Croft, Nick C. Ellis, John Holland,
Jinyun Ke, Diane Larsen-Freeman, Tom Schoenemann
(*Language Learning* 2009, 51: 1-26)



3 Factors and 2 new ideas

- But it is not obvious that **21st century generativism** and **emergentism** are at odds:

EMERGENCE

- ‘the whole is something besides the parts’
(Aristotle *Metaphysics*)
 - It occurs when an entity is observed to have properties its parts do not have on their own, properties or behaviours that emerge only when the parts interact in a wider whole.
 - A familiar refrain in generative linguistics:
 - language = a system which ‘makes infinite use of finite means’ (Humboldt 1836:70)
 - ... ‘whose ‘synthesis creates something that is not present *per se* in any of the associated constituents’ (Humboldt 1836: 67)
- **emergent** properties

3 Factors and 2 new ideas

- Some key properties of **emergent systems** in the sense employed outside of Linguistics (e.g. in the Complex Adaptive/Dynamic Systems literature; Gibb et al. 2019) include:
 - **sensitivity to initial conditions** (path-dependent development > **structural homology**)
 - **self-organization** (“softly assembled” rather than hard-wired properties)
 - **attractors** (biases) and **control parameters** (=cues driving system reorganization/development)
 - **edge of chaos** as ‘**progress niche**’
 - **emergence** (non-linear development where the whole is more than the sum of the parts)
 - **self-similarity** (fractals)

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Early vs late-acquired properties (Tsimpli 2014); macro vs more microparameters
 - **self-organization** (“softly assembled” rather than hard-wired properties) **3 Factors**
 - **attractors** (biases) and **control parameters** (=cues driving system reorganization/development)
Acquisition/Change cues/triggers
 - **edge of chaos** as ‘**progress niche**’ *Coming up!*
 - **emergence** (non-linear development where the whole is more than the sum of the parts)
Humboldt, mappings emerging from **phrase structure** (‘**subject**’, ‘**topic**’), etc.
 - **self-similarity** (fractals)
Parallelisms across **phrase structure** and **phrase types** (lexical core-functional superstructure, phases, etc.)

See **Bosch (2022)** for phenomenal BA dissertation=-length discussion

The third factor

- What is Factor 3?
 - Chomsky (2005: 6)
 - (a) principles of **data analysis** that might be used in language acquisition and other domains;
 - (b) principles of **structural architecture and developmental constraints** that enter into canalization, organic form, and action over a wide range, including principles of **efficient computation**, which would be expected to be of particular significance for computational systems such as language.
- Thinking back to the non-UG factors that have been said to affect language acquisition and diachronic developments, some do have the character of some kind of Factor 3:
 - Transparency
 - Subset Principle (Dell 1981, Berwick 1985)
 - Degree Zero Learnability
 - Least Effort
 - Last Resort
 - For others, this is less clear; they feel like language-specific instantiations of more general constraints / biases:
 - Relativised Minimality
 - Feature Economy
 - Head Preference
 - Late Merge; F*Merge over F*Move
 - Labeling Algorithm
 - Determinacy

(see Van Gelderen (2022) *Third Factors in Language Variation and Change*. CUP)

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(2) **Current minimalist model: a 3 Factors model** (Chomsky 2005)

UG + input + **general cognitive (3rd) factors** → Adult grammar (I-language)

(3) The **Maximise Minimal Means (MMM) model** (Biberauer 2011 *et seq.*)

UG + intake + **MMM** → Adult grammar

where MMM = **a general-cognitive bias** that is active not only in shaping language acquisition, but also i.a.

- **cognitive heuristics and biases** – e.g. Daniel Kahnemann's 'thinking fast' (Kahnemann 2001; cf. also Gigerenzer & Todd's (2000) 'fast and frugal heuristics' and Simon's (1955) bounded rationality/'satisficing', and Ferreira & Patson's (2007) 'good enough' parsing)
- **concept formation** and **cognitive cascades** more generally (Bornstein & Arterberry 2010, Jaspers 2013, Seuren & Jaspers 2014; Rose et al. 2005, 2008, Campos et al. 2000)
- **pattern coding** (centre-surround antagonism) in **vision** (Westheimer 2004) and **gaze heuristics** and **object recognition** (Gigerenzer & Brighton 2009, Rafiei et al. 2021, Leeuwenberg & van der Helm, 2013)
- **human writing systems** (Dehaene 2007, Morin 2018)

The third factor

- What MMM effects might we identify in language?
- Some evidence from within syntax for the role of MMM:
 - crosslinguistically prevalent “**recycling**” **phenomena**
e.g. grammaticalisation, cyclic developments, pragmaticalisation
multi-use [F]s, e.g. [wh], [case], [person], [plural], [dim], [co-in(cide)] (Hale 1986, Wiltschko 2014, Harbour 2020)
“extended” use of early-acquired categories, e.g. consonant/vowel (Toro & colleagues), nominalisation phenomena, V-centred focus (e.g. Immediately Before/After Verb phenomena), etc.
 - (featural) **Relativised Minimality** (Rizzi 1990, 2001, 2013, Starke 2001)
 - structure building via External Merge & Internal Merge (movement) > higher-level duality of patterning (Fortuny 2010) and maybe a connection to Chomsky’s (2004) ‘duality of semantics’
 - Penthouse Principle effects (Ross 1972) & Degree-Zero Learnability (Lightfoot 1989)
- And from language acquisition more generally: “Less is More”/“Start Small” (Newport 1990, Ellman 1993, Yang 2016, Biberauer 2018, Friedemann et al. 2021 *et seq.*, Bosch & Biberauer 2024 *et seq.*) and how keen kids are generalise existing knowledge.

What about Factor 1 (UG) in this model?

- MMM interacts with a bare UG (Merge, Agree, formal feature template), which specifies that grammatical variation be consistently encoded via **formal features** ([F]s), which are distinct from **phonological** ([P]) and **semantic** ([S]) features (Chomsky 1995).
 - UG creates an **[F]-expectation** in respect of grammatical variation.
- Thinking about the **input** an acquirer gets in Chomsky (1995) terms:

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- Thinking about the **input** an acquirer gets in Chomsky (1995) terms:
 - (4) a. lexical sound-meaning mappings: [P]- and [S]-features (**Saussurean arbitrariness**)
 - b. grammatically regulated sound-meaning mappings: [P]-, [S]- and **[F]**-features (**systematic departures from Saussurean arbitrariness**)
 - there are **degrees of arbitrariness** in human language.
 - language-acquiring children evidently pick up on these (recall Wexler (1998)'s 'little inflection machines, and also the Borer-Chomsky Conjecture/BCC approach to minimalist parameters: All parameters of variation are attributable to differences in the features of the functional heads in the Lexicon (Borer 1984:27-29, Chomsky 1995:6, Baker 2008:155f.)
 - these are the **cues** that drive the development of an I-language in child acquisition [and that are not all equally accessible to adult/late learners]

A closer look at Factor 2 (the input)

- Examples of **systematic departures from Saussurean arbitrariness** = **cues** for the acquirer to postulate **formal features (= [F]s)** (Biberauer 2018, 2017/2019)
 - (5) a. **more form than meaning** e.g. concord, doubling phenomena ...
 - b. **no form but meaning**, or *systematic silence* e.g. null arguments, null complementisers, ellipsis, null exponence ...
 - c. **one form with too much meaning**, or multifunctionality, i.e. identical forms whose meanings vary depending on their distribution (“homophony”)
 - d. **high-frequency forms with a very fixed ‘edge’ position** ‘function words’, inflection
 - e. (non-neutral) **movement** (= higher level of duality of patterning) ...

The role of MMM?

- MMM drives acquirers to postulate [F]s sparingly – **Feature Economy** – and to generalise and “recycle” them as much as possible – **Feature Generalization***.
- Acquisition is not just about conservative acquirers (*pace* i.a. William Snyder)
 - the acquirer is **conservative** in postulating (new) features:
 - only where they are **contrastive** (Jakobson 1948, Hall 2007, Dresher 2014, and the Toronto School more generally), i.e. where there is a **cue**, *and*
 - where existing [F]s cannot be ‘recycled’
 - the acquirer is **liberal** about extending existing [F]s to new domains (possibly sometimes in error, and/or to compensate for ambiguous and/or unsystematic input – ‘**going beyond the input**’ scenarios)
- Crucially, MMM also produces a “**Goldilocks effect**” in language acquisition that is familiar from the development of vision and audition (see Kidd, Piantadosi & Aslin 2012, 2014)
 - acquirers systematically attend to input that is neither too simple nor too complex, but ‘just right’
 - At the earliest stages, this is prosody, which allows various kinds of ‘edge’ identification at the pre-lexical stage (see i.a. Gervain & Werker 2008, Biberauer 2018b, 2019a,b for discussion and references)

[* Feature Generalisation is really just the other side of the Feature Economy coin.]

The role of MMM?

- MMM-shaped – so contrast-based, minimal [F]-harnessing – grammar (category) acquisition is predicted to take the kind of shape that is familiar from emergentist work in other domains, e.g. **phonology** (Dresher 2009, via the Successive Division Algorithm (6))

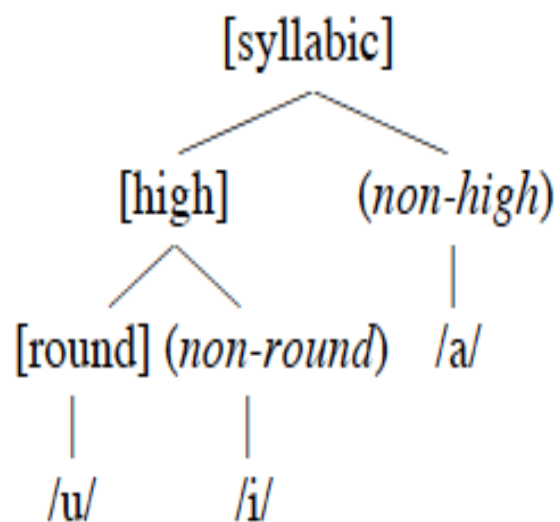
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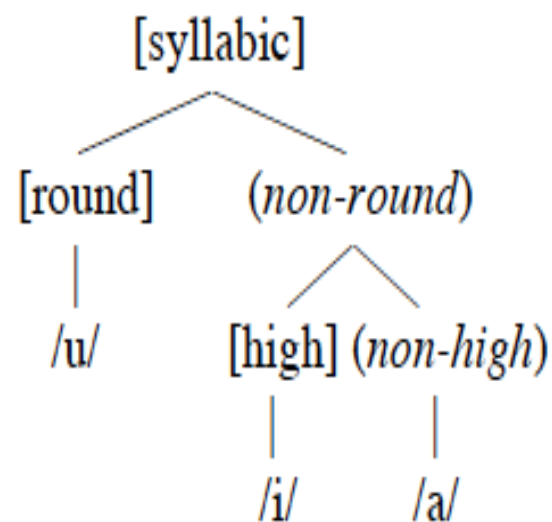
(6) Contrastive hierarchies for /i,a,u/ with [high] and [round]

(from Dresher 2014:168)

a. *[high] > [round]*



b. *[round] > [high]*



- As phonological features are not pre-given, their **hierarchical organization** may vary across languages.
- There appears to be both acquisition and typological evidence for this approach (cf. Fikkert 1994, and also Dresher 2014, Mobbs 2015)

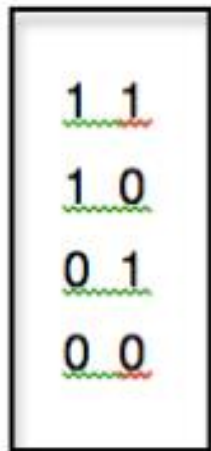
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NB: after the first division, there may be more than one subsequent-division option. Given **structural homology**, each division has consequences for the next > variation emerges readily, but in a systematic way.

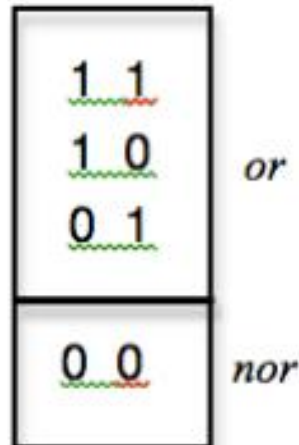
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a. Domain



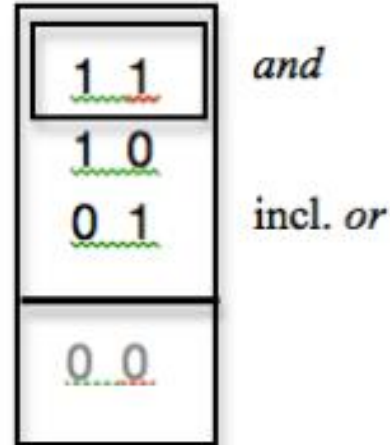
All truth-value pairs

b. Step 1



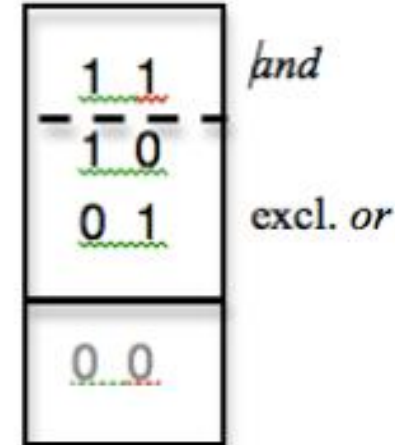
Contradiction
Something is true
vs.
Nothing is true

c. Step 2



Implication
Something is true
vs.
Everything is true

d. Step 2'

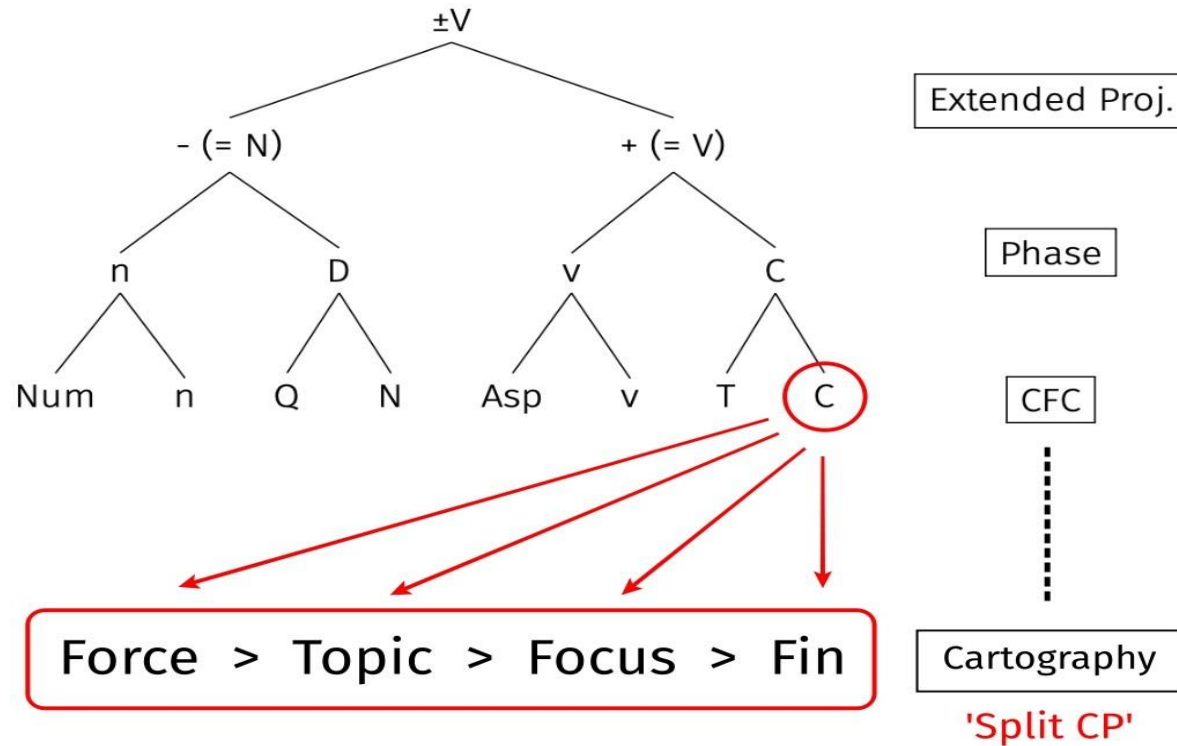


Something but not
everything is true
vs.
Everything is true

MMM shapes category acquisition

- In the domain of syntax, a successive division approach predicts that syntactic category acquisition will exhibit increasing **granularisation**:

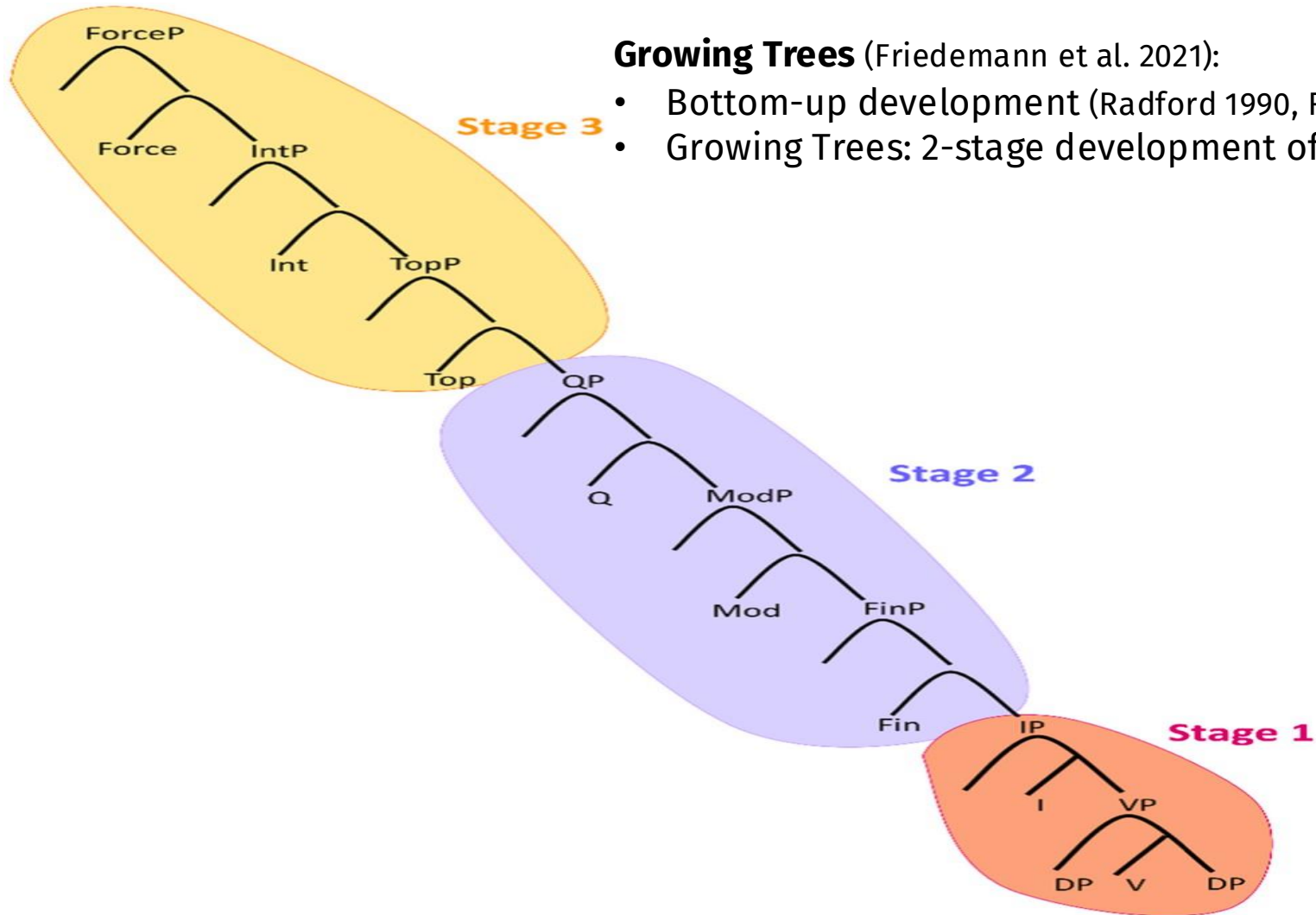
(8) Hypothetical developmental path – an emergent categorial hierarchy (Biberauer & Roberts 2015)



- Structural homology** (=sensitivity to initial conditions): what is acquired early is the foundation for what is acquired later.

MMM shapes category acquisition

- (8) is hypothetical, and it contrasts (significantly!) with what established language-acquisition models have proposed.



Growing Trees (Friedemann et al. 2021):

- Bottom-up development (Radford 1990, Rizzi 1993/4, Diercks et al. 2023): CP last
- Growing Trees: 2-stage development of the articulated CP

MMM shapes category acquisition

- (8) is hypothetical, and it contrasts (significantly!) with what established language-acquisition models have proposed.
- **But** recent and on-going research conducted by Núria Bosch (Cambridge) suggests that a **non-bottom-up granularisation approach** is supported by **developmental patterns in child acquisition**.
 - Studies to date:
 - (i) Bosch (2023a):
illocutionary (= main-clause) complementisers are acquired early
 - (ii) Bosch (2023b, 2025), Bosch & Biberauer (2024):
basic matrix CP > embedded CP and more articulated left periphery (i.e. sub-divided CP)
 - (iii) Bosch & Biberauer (2025a,b)
adjoined/non-moved and “pure operator” topics > “mixed” (non-quantificational, anaphoric) topics
 - (iv) Bosch & Biberauer (2026a,b)
existential expletives > weather expletives

MMM shapes category acquisition

- One study in a little more detail: **Bosch (2023b, 2025), Bosch & Biberauer (2024)**:
 - Key finding: basic matrix CP > embedded CP and more articulated left periphery (i.e. sub-divided CP) (this finding impinges on the Degree-Zero Learnability questions some of you have astutely raised)
 - Method: Longitudinal analysis of 10 typically-developing children in CHILDES, across five languages.

Table 1: Children studied in the CHILDES database and summary information

Language	Corpus	Child	Files	Age	MLUw
Catalan	Serra-Solé	Laura	19	1;07-4;00	1.03-3.47
		Gisela	21	1;07-4;02	1.02-3.51
Italian	Calambrone	Martina	13	1;07-2;07	1.26-2.69
		Rosa	21	1;07-3;03	1.27-3.24
Spanish	Llinàs-Ojea Montes	Irene	59	0;11-3;02	1.0-5.13
		Koki	13	1;07-2;11	1.96-3.61
German	Miller	Kerstin	37	1;03-3;04	1.09-2.89
		Simone	50	1;09-2;09	1.52-4.89
Dutch	Groningen van Kampen	Josse	28	2;0-3;04	1.2-4.01
		Sarah	50	1;06-5;02	1.07-6.07

MMM shapes category acquisition

- One study in a little more detail: **Bosch (2023b, 2025), Bosch & Biberauer (2024):**

■ CP diagnostics:

1. Wh-questions
2. Yes/no questions
(Germanic only)
3. V-to-C movement
(Germanic only)
4. Topics/Foci
5. Illocutionary (main clause) complementisers
(Romance only)
6. Finite embedding

■ Split CP diagnostics (Romance):

1. Top > Wh
2. Top > Top/Foc
3. Complementiser > Wh/Top
4. Quotative *que* 'that' > Wh (Ibero-Romance only)
5. Topic > interrogative *que* 'that' (Catalan only)
6. *Sí que/sì che* 'yes that' and *que sí que* 'that yes that' structures (for the latter, Ibero-Romance only)

MMM shapes category acquisition

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■ Split CP diagnostics (Germanic, V3 orders):

1. Frame-setters
2. Hanging Topic Left-Dislocation
3. Contrastive Left-Dislocation
4. Conditional/temporal clauses with
resumptive *dann/dan* 'then'

MMM shapes category acquisition

- One study in a little more detail: **Bosch (2023b, 2025), Bosch & Biberauer (2024):**

Table 12: Emergence of CP- vs Split CP-structures

	CP-structures	Split CP-structures
Laura	1;10.22 1.15 MLUW	3;03.21 2.54 MLUW
Gisela	2;04.25 1.58 MLUW	2;08.00 2.61 MLUW
Martina	1;08.02 1.57 MLUW	2;04.13 2.69 MLUW
Rosa	1;07.13 1.27 MLUW	2;10.14 2.5 MLUW
Irene	1;04.16 1.32 MLUW	1;11.13 2.95 MLUW
Koki	1;07.20 1.96 MLUW	2;04.18 2.69 MLUW
Kerstin	1;10.03 1.28 MLUW	2;09.11 2.32 MLUW
Simone	1;09.11 1.54 MLUW	2;06.23 2.78 MLUW
Josse	2;00.07 1.2 MLUW	2;11.09 3.57 MLUW
Sarah	1;10.05 1.09 MLUW	3;00.19 3.52 MLUW

MLUw = Mean Length of Utterance
measured in words

For all the children:

- (i) 'basic' CP-structures emerge very early
(see Wexler (1998) on so-called Very Early Parameter
Setting/VEPS; V2 – Very Early)
- (i) 'basic' CP-structures emerge (significantly)
earlier than split-CP structures.

➤ neither early CP nor the 'basic'/'split' split is
predicted by Growing Trees or other models;
it is predicted by the MMM granularisation
model.

MMM shapes category acquisition

- One study in a little more detail: **Bosch (2023b, 2025), Bosch & Biberauer (2024):**
 - Emergence is not just late, but **sudden and ‘explosive’** in the production data ($z = -2.949874$, $p = 0.003$).

Table 3: Production of Split CP-structures before and after MLUw ~ 2.5

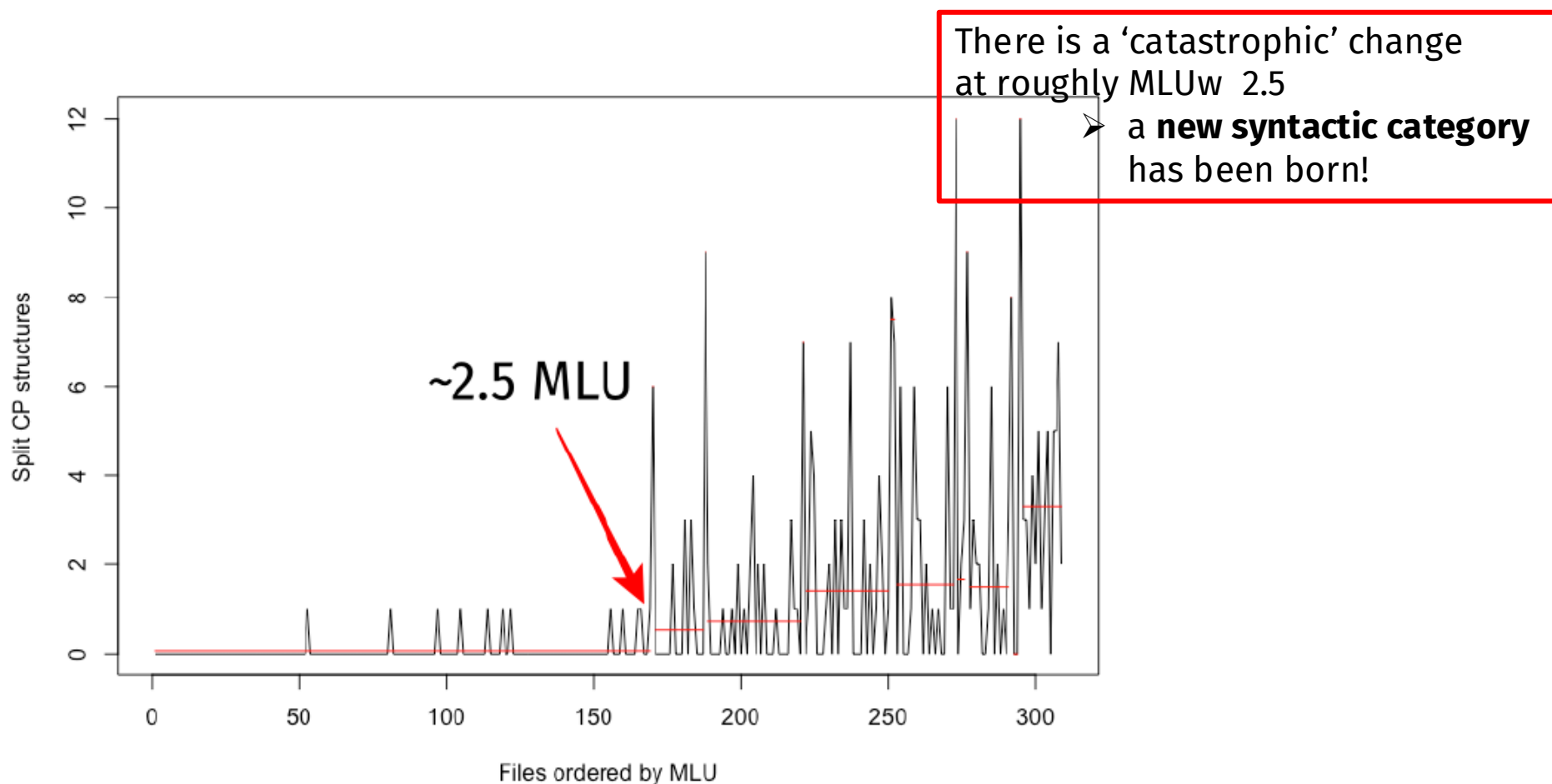
	Before MLUw ~ 2.5	After MLUw ~ 2.5	%
Laura	1	20	4.8-95.2%
Gisela	0	9	0-100%
Martina	0	5	0-100%
Rosa	1	31	3.1-96.9%
Irene	0	85	0-100%
Koki	0	41	0-100 %
Kerstin	3	4	42.9-57.1%
Simone	2	7	22.2-77.8%
Josse	1	19	5-95%
Sarah	2	51	3.8-96.2%
Total	10	272	3.5-96.5%

MMM shapes category acquisition

- One study in a little more detail: **Bosch (2023b, 2025), Bosch & Biberauer (2024):**

- Detecting when the change occurs with change point analysis

(Snyder (2007, 2021) = the inspiration for using change-point analysis)



MMM shapes category acquisition

- In the CP-domain, there is evidence for **refinement** of an early category (C), delivering different split-CP categories.
- Other studies have shown similar refinements in other domains:
 - De Lisser et al. (2017) on the acquisition of the TMA field in Jamaican Creole;
 - Sanfelici & Gallina (2022) and Mitrofanova (2018) on the development of (complex) PPs;
 - Heim & Wiltschko (2025) on the development of interactional language (at the very top of the clause, above CP, so also not predicted by the Growing Trees approach)
- Syntactic categories seem to granularise, as predicted by the MMM approach.
- The categories appear to be emergent and the specific CP-internal and other domain-internal distinctions that emerge are likely to vary by language, depending on the nature of the systematic departures from Saussurean arbitrariness that serve as the cue to postulate a new [F] to split the original category (which thus turns out to have been underspecified in respect of grammatical distinctions that are made – and systematically signalled in the input – in the relevant system).
- The greater variation between fine-grained categories is expected on an emergentist generative approach [ever more so the more specialized (i.e. richly featurally specified) the category becomes]; it is problematic on a UG-centred approach (at least in the 3 Factors, poor-UG era).

MMM shapes category acquisition

- Before moving on: what this study showed about the acquisition of embedded CP
 - Recall Lightfoot's (1999) Degree-Zero Learnability.
 - Leaving aside the 2 outliers (Koki is generally an outlier) and focusing on MLUw, rather than age as before, embedding consistently develops **after** topicalisation, which was one of the 'basic' CP-diagnostics (when it occurs alone).
 - Why this is potentially interesting from a Degree-Zero perspective:
 - The matrix-first intuition seems correct for the category C > children may start out being Degree-Zero learners as this is initially neither too simple nor too complex, but 'just right' (Goldilocks)
 - When they acquire embedded Cs (e.g. complementisers), they will already have various features and (variably specified) categories at their disposal to "recycle" in embedded contexts, and, potentially, to further elaborate.
 - *expectation*: a more nuanced Rossian Penthouse House effect; and that acquisition studies could be very illuminating!

Table 3 Emergence of topicalisation vs embedding markers

	Topicalisation	Embedding
Laura	2;08.03 1.88 MLUw	3;00.02 2.42 MLUw
Gisela	2;08.00 2.61 MLUw	2;08.00 (same file) 2.61 MLUw
Martina	1;08.17 1.56 MLUw	1;11.20 1.99 MLUw
Rosa	2;04.29 1.77 MLUw	2;06.29 2.6 MLUw
Irene	1;08.09b 2.24 MLUw	1;09.10 3.28 MLUw
Koki	1;11.25 2.47 MLUw	1;11.25 (same file) 2.47 MLUw
Kerstin	2;00.05 1.76 MLUw	2;07.23 2.13 MLUw
Simone	1;10.20 1.62 MLUw	2;04.20 1.96 MLUw
Josse	2;03.28 1.94 MLUw	2;09.02 2.42 MLUw
Sarah	2;00.17 1.68 MLUw	3;00.19 3.52 MLUw
Average	1.93 MLUw	2.54 MLUw

MMM shapes category acquisition

- Where does this leave us in relation to more “classic” questions in diachronic syntax, e.g. word-order changes (OV > VO, and the reverse; loss of V2, V-to-T, etc.)?
- **MMM and OV/VO:**
 - The ‘basic’ fixing of OV/VO is known to be early (it’s another of Wexler’s VEPs).
 - There is a lot of research suggesting that the **basic** OV/VO pattern is fixed during the pre-lexical stage, even pre-birth (see i.a. Gervain & Werker 2008 and Dye et al. 2019 for overviews)
 - Crucially, ‘basic OV/VO’ here seems to mean the bottommost head-complement pair within the clause (i.e. V and O, or the head-complement pair in the domain to which **nuclear stress** – the primary pitch accent (the loudest, longest syllable) – is assigned; see Cinque 1993).
 - The striking OV-postposition/VO-preposition correlation highlighted in Baker (2008) as evidence in favour of macroparameters actually only relates to V and P; not to all heads in the system (recall the OV typology from Day 1)

- (9)
- | | | |
|------------------------|-----|---|
| OV and Po(stpositions) | 472 | Postpositions often turn out to be case markers (=reflexes of case) rather than case assigners (like P); so the higher number there deserves caution. |
| OV and Pr(epositions) | 14 | |
| VO and Po | 42 | |
| VO and Pr | 456 | |
- From Baker (2008)
- The correlation may reflect the strength of the initial prosodic cue in fixing the directionality of the prosodically strongest elements in the speech stream.
 - Subsequent categorial differentiation may or may not retain **the directionality profile** of the bottommost pair (harmonic vs disharmonic systems)
 - **Structural homology** ensures that successive divisions build on the established properties of prior (early-acquired, less formally differentiated) categories > harmony effects (including FOFC).

MMM shapes category acquisition

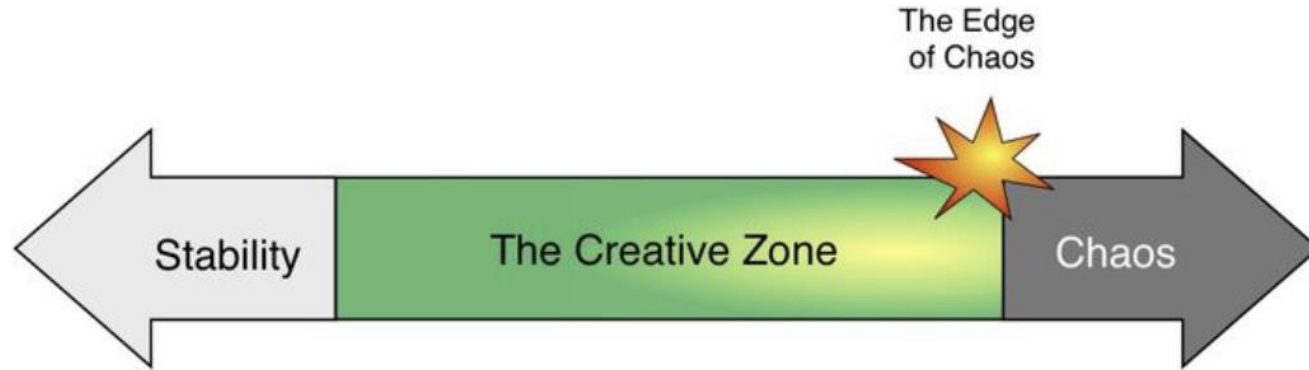
- Where does this leave us in relation to more “classic” questions in diachronic syntax, e.g. word-order changes (OV > VO, and the reverse; loss of V2, V-to-T, etc.)?
- **MMM and OV/VO, and V-to-C, and V-to-T, etc:**
 - The increasing granularisation approach to categorial acquisition outlined here also leads us to expect the kind of typological picture that has emerged from research into the forms that OV and VO systems can take
 - macro > meso > micro > nano-sized OV and VO systems (Biberauer & Roberts 2012 *et seq.*)
 - The same expectation holds for other phenomena that we previously treated parametrically:
 - V2 can be identified crosslinguistically as a C-level phenomenon (a mesoparameter for Biberauer & Roberts 2016a,b), but individual V2 languages seem to differ as to the precise nature of the C-head that the finite verb moves to (e.g. Force, Fin or, as in modern English, Foc)
 - V-to-C loss – either across the board for an entire language or in a particular grammatical domain, as in Conditional Inversion/CI in the history of English (Biberauer & Roberts 2016a,b) – is expected to involve successive narrowing of the class of contexts (across-the-board loss) or verb-types (CI loss) targeted (see Biberauer & Roberts 2016a,b for discussion of CI)
 - V-to-T loss is likewise expected to take place via smaller steps, which it evidently did in the history of English (see Han 2000, Haeberli & Ihsane 2016).

MMM shapes category acquisition

- Where does this leave us in relation to more “classic” questions in diachronic syntax, e.g. word-order changes (OV > VO, and the reverse; loss of V2, V-to-T, etc.)?
- **MMM and OV/VO:**
 - The ‘basic’ fixing of OV/VO is known to be early (it’s another of Wexler’s VEPs).
 - If OV/VO is fixed early (first) and categorial differentiation (granularisation) follows along the lines of what was outlined here, we might also be able to understand why OV > VO change seems more readily possible than VO > OV change.
 - FOFC shows us that VO > OV change must start at the bottom; starting at the top/above VP would give a FOFC-violating structure – e.g. VO-T/VO-C.
 - Starting at the bottom would require a significant disruption of the prosodic profile, however, as this is fixed early, i.e. we expect this to be more likely in heavy-contact situations.
 - OV > VO change, by contrast, has to proceed top-down; starting at the bottom would again produce a FOFC-violating order – e.g. VO-T or VO-C.
 - The combined fact that innovating (e.g. via reanalysis/rebracketing, as often assumed for *that*-type complementisers) or borrowing (integrating; initially, via adjunction at the edge; see discussion to follow) initial C-elements seems relatively simple and that the basic prosodic profile can remain unchanged would seem to favour OV < VO change.
 - NB: This does **not** mean that VO > OV change is ruled out; it is just predicted to require more significant perturbation of the input children receive.

Another emergent phenomenon

- Recall the **edge of chaos** (a phenomenon associated with emergent systems)

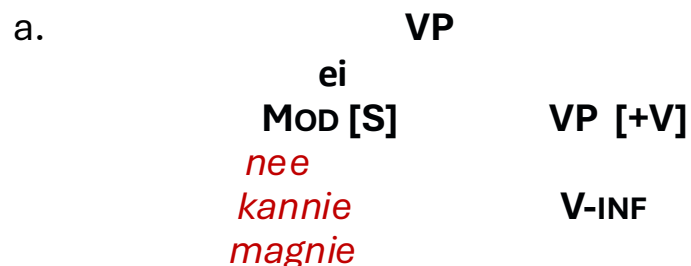


- Edge-of-chaos** = **creative zone** ('progress niche') where progress/creativity happens
- But note that **progress/creativity** builds on and **develops already-established structure**
 - **structural homology**
- Maximise the available means**
 - 2 means that are always available in a grammatical system with some already-acquired structure
 - (defined by features [F]s in current theory)
 - (i) the **[F]-defined structure** and
 - (ii) the [F]-defined structure's **edge**.

The edge of chaos in acquisition

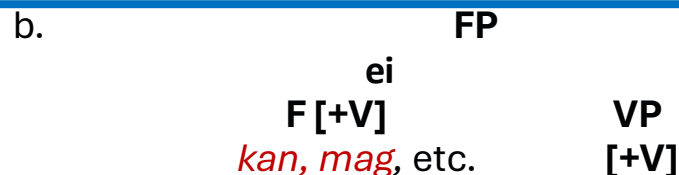
- the significance of the **EDGE**: prosodically mediated pre-lexical ability to distinguish content and function words (Shi, Werker & colleagues).

(9) **Hoekstra & Jordens (1996)**: The acquisition of the Dutch modal system (Root Infinitive stage)



(MOD = **semantic** category of negative modal elements; ≠ [+V])

- Syntactic structure of the modal system (Jasmijn at 1;10)
 - doubly stressed adjunction pattern: *kánnie zíttēn* ('cannot sit')
 - kan*, *mag*, etc. never used without *nie*; complementary distribution with *nee* (=anaphoric negator)



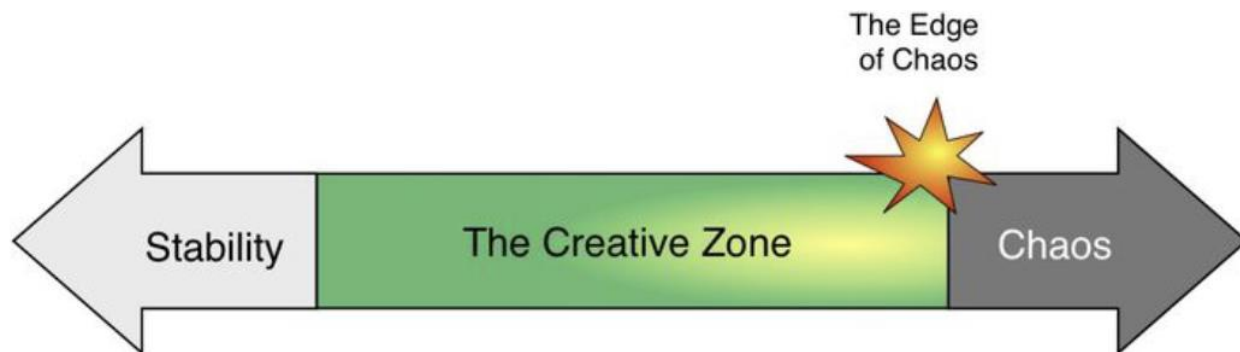
- Syntactic structure of the modal system (Jasmijn at 1;11)
 - integrative stress pattern: *kannie zíttēn*
 - kan*, *mag*, etc. start to appear independently of *nie*. Modals recognised as [+V], i.e. the [F] initially ascribed to lexical verbs is extended to the class of modals, and negation markers are separated out from this class (they're not [+V]).

The edge of chaos beyond acquisition

- In **acquisition**, **edge effects** of this kind seem common, particularly at the earliest stages of grammar construction (see much work since Klima & Bellugi (1966, Bellugi 1967) by Lebeaux, van Kampen, Roeper, a.o., and recent discussion i.a. in Diercks et al's (2023) *Developmental Minimalist Syntax*, the work of Bosch, Heim & Wiltschko 2025)
 - a key distinction in acquisition: emergence (= initial integration) vs mastery (full grammatical integration)
- There are indications, though, that the same **initial-adjunction route** also characterizes what happens where language variation arises, i.e. that the 'edge of chaos' is also significant in **variation** and **change**, including in the innovative output of adults.
 - adjunction (without [F]s) may also precede projection and (full) grammaticalization (i.e. [F]s)
 - we see many **innovative edge effects** in language

The edge of chaos beyond acquisition

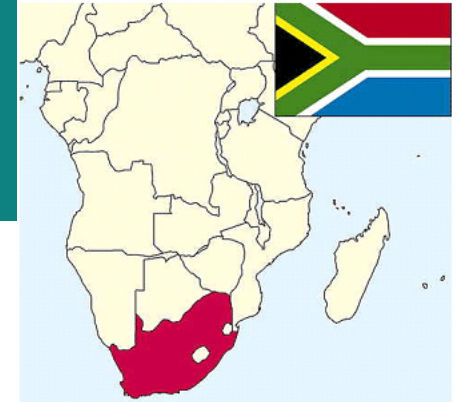
- Proposal: The “**way in**” for new material – in **acquisition** as in **contact** and **change** generally – is via the **edge of a structural domain**.



- **STABILITY** = [F]-defined structure (the stable grammar)
- The **EDGE** = not [F]-defined, i.e. a zone available for the **adjunction** of new elements

- What **edges** can we identify in syntax?
- **Phase edges, or Layered Derivation** (Zwart 2011) **edges**
 - If **Lexical Arrays** define **phases**, phase edges = the “way in”: elements lacking formal features [F]s cannot select or be selected > “last out” of their Lexical Array (Biberauer 2017)
 - Prediction: roots ([F]-less) can be integrated by merging with a categorizer [$v > [V]$, $n > [N]$, etc., as in DM) **or** at a structural (phase) edge.
 - Expectation: “new” (e.g. “recycled”, borrowed, or newly created) roots being integrated as new content words and also at the edges of structural domains (e.g. vP, CP, DP, etc.)

The edge of chaos beyond acquisition



- A case study: the rise of the Afrikaans sentence-final concord element *nie*₂

- In contrast to Dutch, Afrikaans is a Negative Concord language:

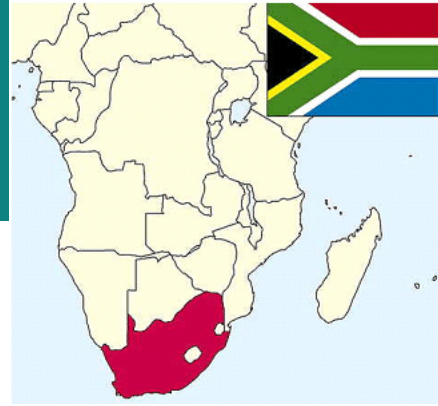
(10) Hulle is **nie**₁ moeg **nie**₂. (Dutch: Ze zijn **niet** moe.)
they is not tired NEG
'They are not tired.'

- *nie* plausibly originates in resumptive tag-containing structures like (11) (Roberge 2000, Biberauer 2015)

(11) a. Het kan niet waar zijn, **nee**! b. Jij komt niet mee, **ne**?
it can not true be no you come not with PRT
'It can't be true, no!' 'You're not coming along, right?' [expectation: negative answer]

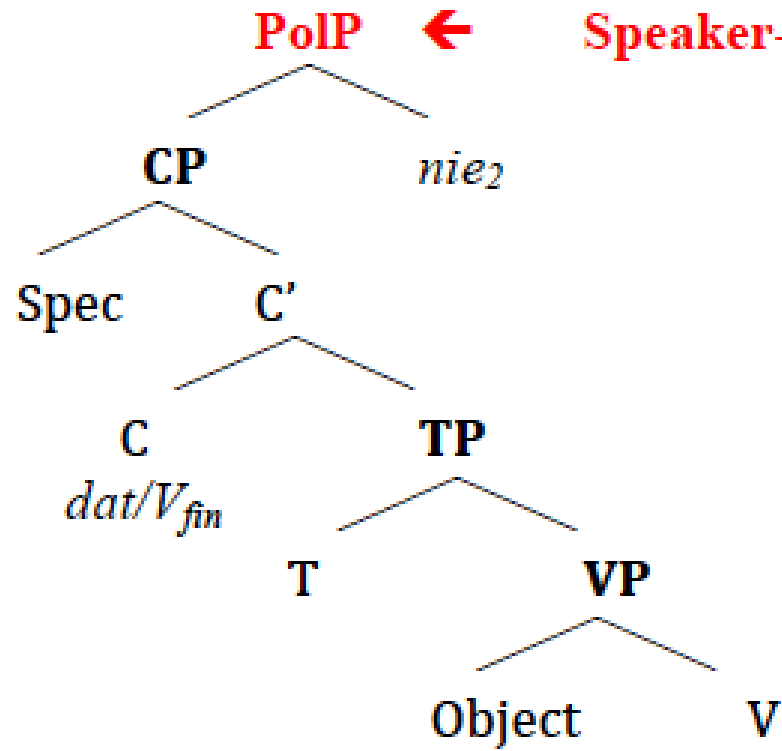
- *ne* = [F]-less particle; therefore we expect it to be merged at the edge
 - anaphoric negator *nee* may bear [neg], but not a selecting [F] that would account for its peripheral integration in (11a).
- the tag was integrated via **adjunction**, i.e. not on an [F]-regulated basis, at the edge of the clause (CP).

The edge of chaos beyond acquisition



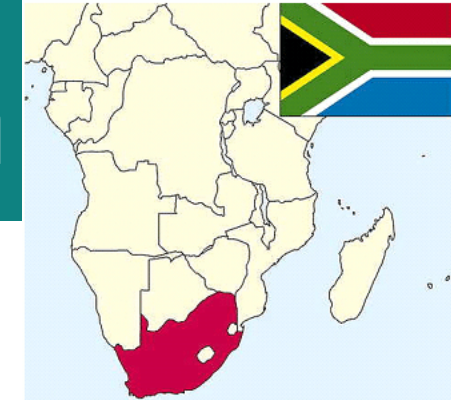
- A case study: the rise of the Afrikaans sentence-final concord element *nie*₂
 - The result: the **grammaticalisation** of the **initially emphatic** *nie*₂ led to a **new structural layer above CP, PolP**.

(12)



- Standard West Germanic clauses = **CPs** with finite V/subordinator in C
- Modern Afrikaans negative clauses = **PolPs**
 - initially: Pol = externally adjoined adjunct plausibly introduced into the input by **adults**
 - frequent use leads a generation of acquirers to postulate a grammaticalised (i.e. [F]-bearing) peripheral projection, PolP, in negative clauses.
 - This PolP was subsequently extended to all XP-edges

The edge of chaos beyond acquisition



- MMM drives acquirers (and maybe adults too?) to **extend** this innovated peripheral option.

➤ optional final *nie* at the edge of other XPs:

(13) a. Dit is [nie die GELD (**nie**)], maar die TYD wat hom pla.
it is not the money POL but the time what him bug
'It's not the MONEY, but the TIME that's bothering him.'

[DP-edge]

b. Dit moet [nie langer (**nie**)] as 10 000 woorde wees.
it must not longer POL than 10 000 words be
'It mustn't be ANY longer than 10 000 words.'

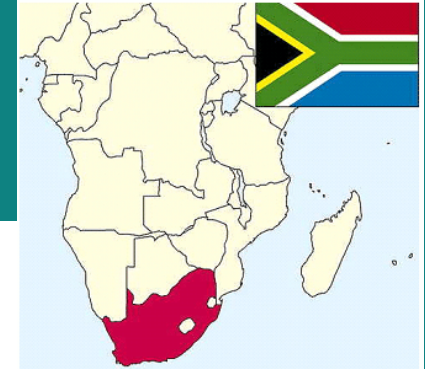
[AP-edge]

c. Hy is [niks (**nie**)] tevrede nie.
he is nothing POL satisfied POL
'He is not remotely satisfied.'

[negative DP-edge]

- **expressivity** at the edge of sub-clausal XPs (optionality > additional interpretive effect – *Fox-Reinhart* pattern)
- Biberauer (2018 *et seq.*): structural edges = more generally a domain where we see expressively oriented recycling; this is particularly evident in spoken varieties, strikingly so in those used in complex sociolinguistic circumstances.
- edge innovation is a structural pattern we might expect to see in the grammatical creativity of both kids and adults.

The edge of chaos beyond acquisition



- Some further examples of edge-oriented innovation in Afrikaans:
 - Differential Object Marking (DOM) (Molnárfi 1999, 2001, Biberauer 2018, 2026):
- Kaaps and Colloquial Afrikaanses more generally have innovated **Differential Object Marking (DOM)** with *vi/vir* ('for'):

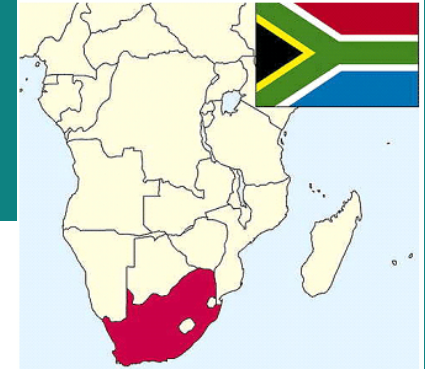
(14) ... hy't geTRY om **vi my en my ma** te skiet. [Kaaps]
he'd tried INF.C for me and my mom to shoot
'... he tried to shoot me and my mom.' (Spoekie, p.106)

(15) ... hoo ek **vi Antie Sophie** inkom
hear I for Auntie Sophie in.come
'... I hear Auntie Sophie coming in.'

(16) Hulle het **vir daardie Jack Daniels** geklits!
they have for that Jack Daniels beat.PRT
'They really murdered that Jack Daniels!'

- **optional marking**, particularly common with **human, animate** and **specific** objects; extends beyond **affected entities** (15) and to **affectively marked inanimates** (16)

The edge of chaos beyond acquisition



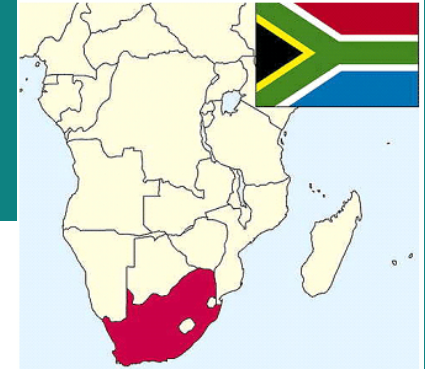
- Some further examples of edge-oriented innovation in Afrikaans:

- Affective adjectival –e (partly discussed by Lass 1990 as an instance of linguistic **exaptation**)

- (18)
- a. ‘n swaar tas / las
a heavy bag burden
‘a heavy bag/load’ (literal)
- b. ‘n sware las
a heavy-E burden
‘a heavy burden’ (figurative)
- c. Dis ‘n sware ou tas!
this.is a heavy-E old bag
‘This is really a jolly heavy bag!’ (affective)

- –e ion is regulated by gender
- Its distribution is heavily restricted, and even Dutch L1-acquiring children only acquire it late (Tsimpli & Hulk 2013)
- Gender was lost very early in the history of Afrikaans, rendering the –e inflection “junk” – it could just have been eliminated.
- Instead, modern Afrikaans incorporates it both prosodically (multisyllabic adjectives require it) and, crucially, expressively, as in (18).
- In MMM terms, (18) is just another instance of edge innovation (pragmaticalisation; see Corver 2016 on harnessing of sub-morphemic elements for expressive purposes)

The edge of chaos beyond acquisition



- Some further examples of edge-oriented innovation in Afrikaans:

- Quirky Verb Second (de Vos 2006)

(19) a. **Waar kom haal** jy ons?

where come fetch you us

‘Where will you come and fetch us?’

[regular V2: **Waar haal** jy ons?]

b. **Wanneer loop koop** jy my roomys?

when walk buy you my ice-cream

‘When will you go (not necessarily on foot) and get my ice-cream?’

[regular V2: **Wanneer koop** jy my roomys?]

- V_1 = a semi-grammaticalized motion/posture verb (Biberauer 2017, Cavirani-Pots 2020)
- V_2 = a lexical verb, which is modified by V_1

➤ more expressive (perspectival/stance-oriented) meaning than the unmodified V_2

- The (inter)subjective interpretations at stake here and in the other optional edge cases discussed are key to understanding the grammatical developments often referred to as **pragmaticalisation** (see i.a. Diewald 2011, Müller & Axel-Tober 2025, Mosegaard Hansen & Waltereit 2025)

Concluding thoughts

Concluding thoughts

- For diachronically oriented generativists, there has been the sense from very early on that two factors are not enough.
- Additionally, diachronically oriented work has underlined the difficulties associated with a model where cues/triggers/p-expressions need to map onto UG-specified content.
 - The 2-Factor approach always lacked a learning theory and never resolved the Linking Problem.
- The 3 Factors approach articulated in Chomsky (2005) seems very promising indeed, not only when it comes to addressing the Linking Problem (which doesn't arise; there is no specific UG content to link to) that was clear in the (synchronic) acquisition context, but also when it comes to linking the '3 Factor precursors' from diachronic work to a plausible cognitive entity.
- MMM, as a 3rd Factor of the kind envisaged by Chomsky (2005), seems to facilitate insight not only into broad questions like some of the ways in which linguistic categories are like cognition-imposed categories more generally, but also into the respects in which recurring properties of grammars have emergent properties, which seem to underlie many of the patterns that generative research, synchronic and diachronic, has uncovered.

Thank you!

Baie dankie!